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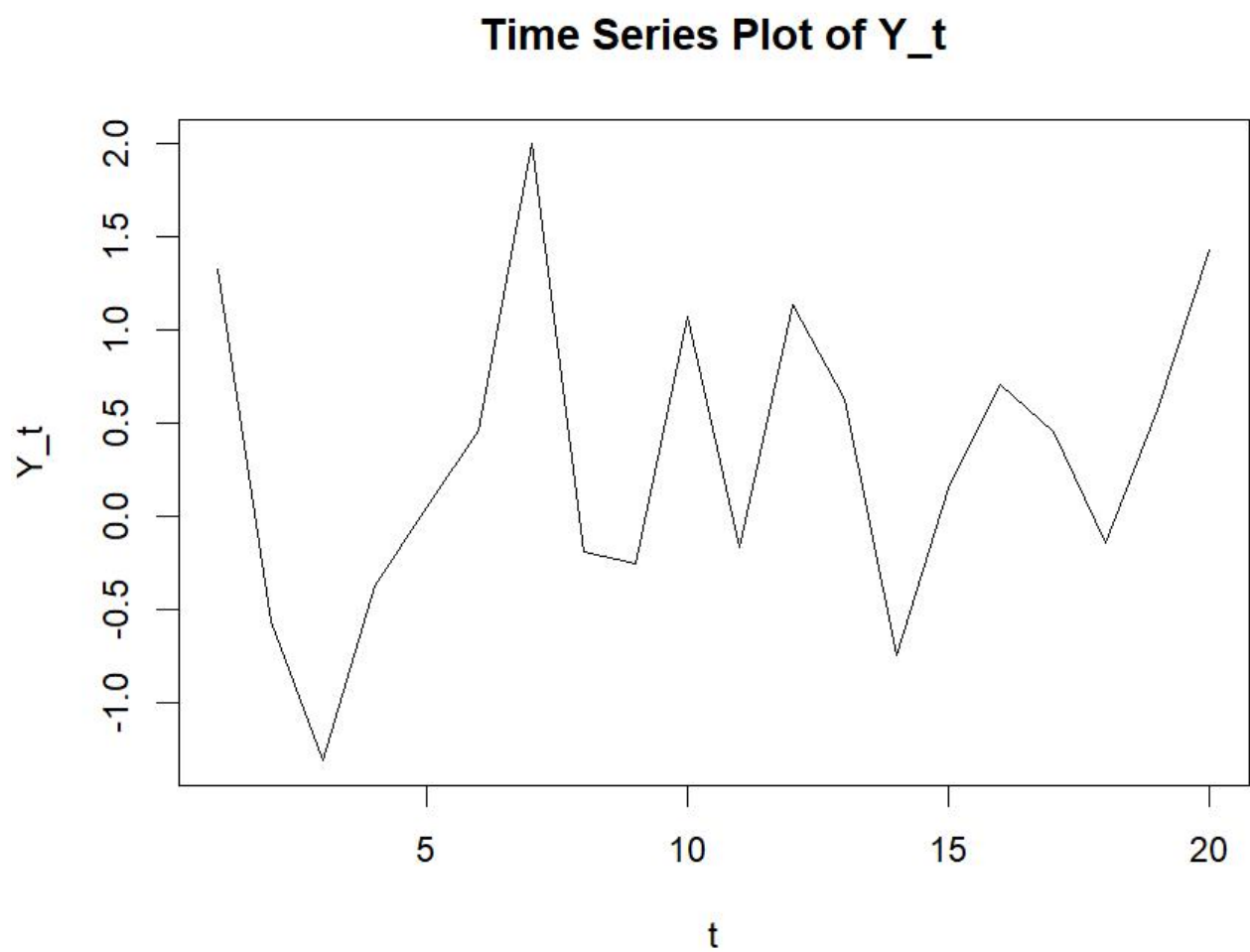
SID: 1155211130

STAT 4005 ASG3

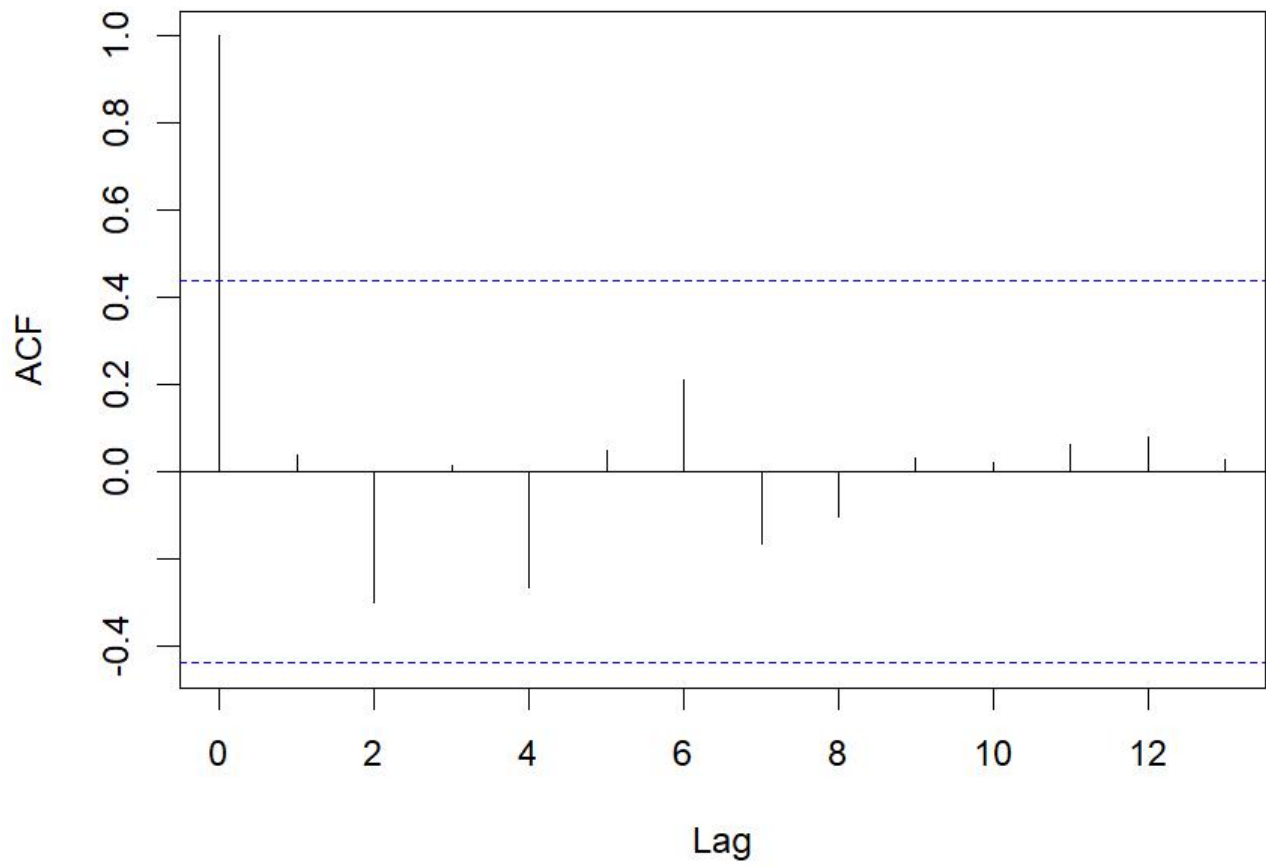
All R code are in the `DaiDasen_ASG3.R`

Problem 1

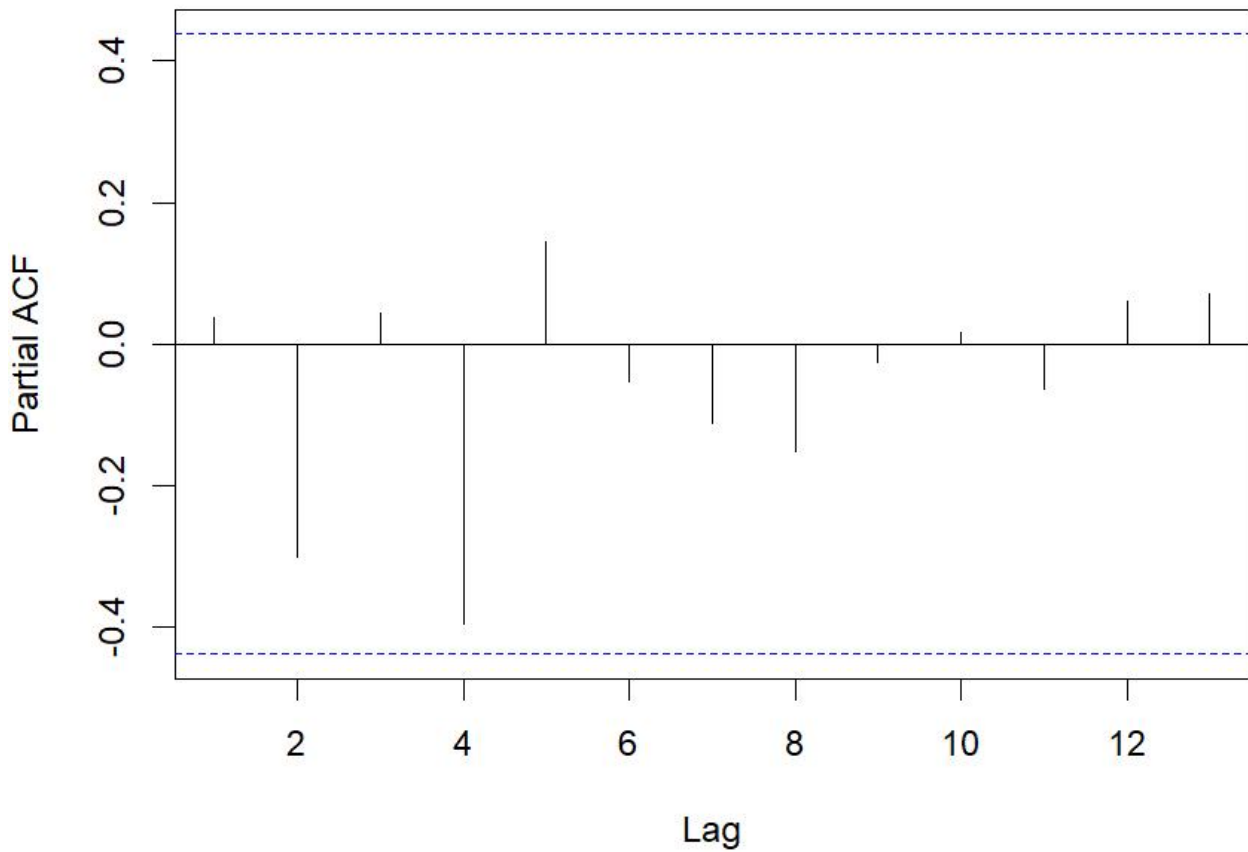
The time series plot, ACF plot, and PACF plot are as follow



Sample ACF of Y_t



Sample PACF of Y_t



Problem2

For the MA(1) model $Y_t = Z_t + \theta Z_{t-1}$, the method of moments estimates are:

$$\hat{\theta}_{moment} = 0.0362849, \hat{\sigma}_{moment}^2 = 0.6414742$$

Problem 3

For the AR(2) model $Y_t - \phi_1 Y_{t-1} - \phi_2 Y_{t-2} = Z_t$, the Ordinary Least Squares estimates are: $\hat{\phi}_1$ (LS) = 0.2353662, $\hat{\phi}_2$ (LS) = -0.2230385, $\hat{\sigma}^2$ (LS) = 0.7236343

The 95% confidence intervals for the coefficients were:

$$\text{95\% CI for } \phi_1: [-0.2301156, 0.700848] \quad \text{95\% CI for } \phi_2: [-0.6885203, 0.2424433]$$

Problem 4

The Yule-Walker estimates for the AR(2) model parameters are:

- $\hat{\phi}_{1\ YW} = 0.04718125$
- $\hat{\phi}_{2\ YW} = -0.3020055$

Problem 5

For the ARMA(1,1) model $Y_t - \phi Y_{t-1} = Z_t + \theta Z_{t-1}$, the Conditional Least Squares estimates are:

- $\hat{\phi}_{\text{CLS}} = -0.5108904$
- $\hat{\theta}_{\text{CLS}} = 0.8439972$
- $\hat{\sigma}_{\text{CLS}}^2 = 0.6469631$

Problem 6

The Maximum Likelihood (ML) estimates for the ARMA(1,1) model are:

- $\hat{\phi}_{\text{ML}} = -0.3882917$
- $\hat{\theta}_{\text{ML}} = 0.9999999$
- $\hat{\sigma}_{\text{ML}}^2 = 0.5253896$

The maximized value of the log-likelihood for this model is **-23.08444**.

Problem 7

AR(p) models for $p = 1, \dots, 5$ were fitted, and their Final Prediction Error (FPE) was calculated.

P	1	2	3	4	5
FPE	0.80	0.84	0.89	0.93	0.85

The model with the minimum FPE is AR(1)

Problem 8

MA(q) models for $q = 1, \dots, 5$ were fitted, and their Corrected Akaike Information Criterion (AICC) was calculated.

Q	1	2	3	4	5
AICC	53.38	53.20	56.28	59.42	63.60

The model with the minimum AICC is MA(2).

Problem 9

Residual Matrix

```
Time Series:
Start = 1
End = 20
Frequency = 1
[1] 9.404523e-01 -1.000209e+00 -4.272393e-01 5.694935e-14
[5] 4.564356e-02 3.873015e-01 1.535416e+00 -1.533244e+00
[9] 1.134204e+00 -5.720777e-03 -1.575403e-01 1.240193e+00
[13] -5.411121e-01 -2.208210e-01 3.517961e-01 3.583465e-01
[17] 9.946920e-02 -2.303545e-01 7.741009e-01 6.592207e-01
```

Ljung–Box test $Q(10)$:

- Statistic: $Q(10) = 11.73$
- $df = 9$
- $p\text{-value} = 0.2289$

Hypotheses:

- H_0 : residuals are white noise (no autocorrelation up to lag 10).
- H_1 : residuals have autocorrelation at some lag ≤ 10 .

Conclusion ($\alpha = 0.05$):

- $p\text{-value} (= 0.2289 > 0.05) \rightarrow$ **fail to reject (H_0)**.
- Residuals are consistent with white noise; MA(1) fit is adequate.

Problem 10

Combining the results:

- Best AR model by FPE: $AR(1)$
- Best MA model by AICC: $MA(2)$
- $MA(1)$ residuals look like white noise but has larger AICC than $MA(2)$.

Since AICC directly compares model fit and complexity and is smallest for $MA(2)$, we would select **the $MA(2)$ model** to describe the data.